

SIMATS ENGINEERING

ASSESSMENT TOOL 1

**COURSE NAME: Artificial Intelligence
CSA17**

COURSE CODE:

COURSE OUTCOME COVERED

CO5: *Design AI-based systems using PySpark, RDD operations, and intelligent agent architectures, using scalable systems and integrate components in distributed AI applications. (BTL 5)*

Assessment Tool Weightage: 50%

QUESTIONS: 5

Annexure A

**Duration : 1 Hour
Total Marks:50**

Design Task

Code of Conduct:

*I POOJASREE B (192410091) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: POOJASREE B

Design Task

AI-Powered Smart Home Controller System

Design a scalable AI-powered smart home controller that uses PySpark RDDs to process sensor data streams (temperature, motion, energy usage) from thousands of connected homes. Explain how RDD transformations support large-scale distributed processing of this sensor data, and describe how intelligent agents within each home coordinate to optimize comfort, energy efficiency, and security while adapting to occupant behavior patterns.

AI-Powered Smart Home Controller System

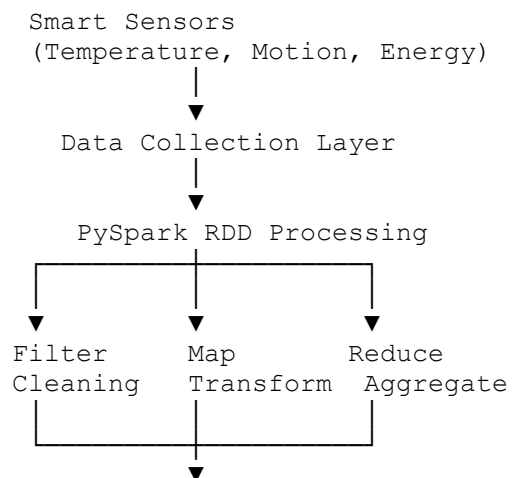
1. System Overview

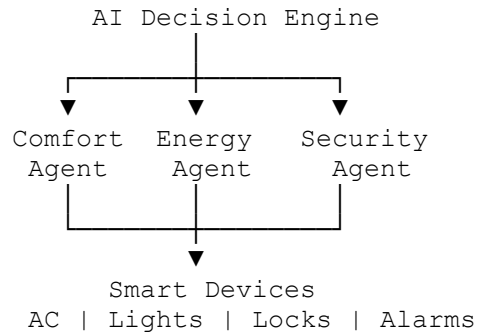
An **AI-Powered Smart Home Controller System** is designed to manage and automate smart devices across thousands of connected homes. The system collects sensor data such as **temperature, motion detection, and energy consumption** and processes it using **PySpark RDDs** for scalable distributed computation.

Intelligent agents within each home analyze the processed data and make autonomous decisions to improve:

- **Comfort** – maintaining preferred temperature and lighting.
- **Energy Efficiency** – reducing unnecessary power consumption.
- **Security** – detecting unusual motion or unauthorized activity.
- **Personalization** – adapting to occupant behavior patterns.

2. System Architecture





3. Sensor Data Processing Using PySpark RDDs

Each sensor generates continuous data records. A sample sensor record may contain:

```
(Home_ID, Sensor_Type, Value, Timestamp)
```

Example:

```
(H101, Temperature, 32.5, 10:30 AM)
(H101, Motion, Detected, 10:31 AM)
(H101, Energy, 4.2 kWh, 10:32 AM)
```

PySpark distributes this data across multiple computing nodes using **Resilient Distributed Datasets (RDDs)**.

RDD Transformations

a) `map()`

Used to transform raw sensor data into a structured format.

```
sensor_rdd.map(lambda x: x.split(","))
```

It converts raw sensor records into individual fields for further processing.

b) `filter()`

Used to select important or abnormal sensor readings.

```
temperature_rdd.filter(lambda x: float(x[2]) > 30)
```

This identifies homes with high temperature levels.

c) `reduceByKey()`

Used to aggregate sensor readings from multiple homes.

```
energy_rdd.reduceByKey(lambda a, b: a + b)
```

This calculates total energy consumption efficiently across distributed nodes.

d) `groupByKey()`

Groups sensor readings belonging to the same home.

```
rdd.map(lambda x: (x[0], x)).groupByKey()
```

This helps analyze all sensor activities for a particular home.

e) `flatMap()`

Useful when processing multiple events or patterns from a single sensor record.

These transformations are executed in parallel across a cluster, making the system scalable for thousands or millions of sensor events.

4. Intelligent Agents in Each Home

Each smart home contains multiple autonomous AI agents.

Comfort Agent

The Comfort Agent monitors temperature and occupancy.

Actions:

- Adjusts air conditioner temperature.

- Controls fans and heaters.
- Adjusts lighting based on occupancy.
- Learns preferred environmental settings.

Example:

High Temperature + Occupant Present

↓

Comfort Agent

↓

Turn ON AC

Energy Management Agent

The Energy Agent monitors electricity usage.

Actions:

- Identifies high-energy-consuming devices.
- Switches off unused appliances.
- Schedules appliances during low-demand periods.
- Optimizes energy consumption.

Example:

No Motion Detected + Lights ON

↓

Energy Agent

↓

Switch OFF Lights

Security Agent

The Security Agent analyzes motion and activity patterns.

Actions:

- Detects unusual movement.

- Activates cameras.
- Locks doors automatically.
- Sends alerts to homeowners.

Example:

Motion Detected at Unusual Time



Security Agent



Camera Activation + Alert

5. Coordination Between Intelligent Agents

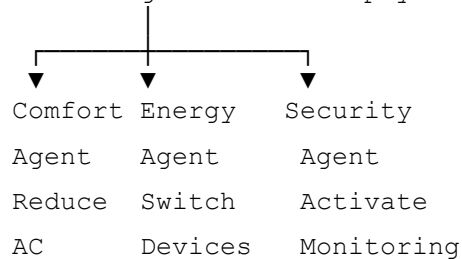
The agents communicate and coordinate their decisions to avoid conflicts.

For example:

- The **Motion Agent** detects that nobody is at home.
- The **Comfort Agent** reduces air conditioning.
- The **Energy Agent** switches off unnecessary appliances.
- The **Security Agent** activates security monitoring.

This coordination creates a cooperative multi-agent system.

Motion Agent → Home Empty



6. Adapting to Occupant Behavior

The AI system continuously learns occupant behavior patterns using historical sensor data.

For example:

Observed Pattern	Learned Action
User arrives at 6 PM	Pre-cool the room
No movement for 30 minutes	Turn off lights
High energy usage at night	Optimize appliance usage
Unusual midnight motion	Trigger security alert

Machine learning algorithms can identify:

- Daily routines.
- Preferred temperatures.
- Typical occupancy times.
- Appliance usage patterns.
- Normal and abnormal movement patterns.

Over time, the system becomes more personalized and requires less manual control.

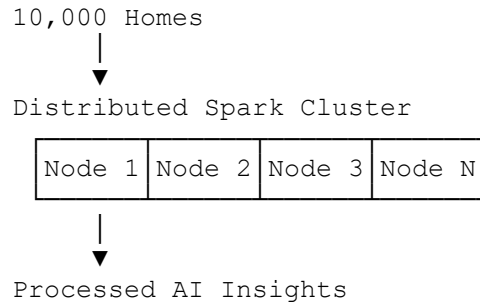
7. Scalability of the System

PySpark RDDs make the system scalable because data is distributed across multiple machines.

Scalability Features

- Parallel processing of sensor streams.
- Fault tolerance through RDD lineage.
- Distributed storage and computation.
- Ability to process data from thousands of homes.
- Efficient aggregation using `reduceByKey()`.
- Horizontal scaling by adding more worker nodes.

For example, instead of processing data from 10,000 homes on one server, PySpark distributes the workload across multiple cluster nodes.



8. Conclusion

The AI-Powered Smart Home Controller combines **PySpark RDD-based distributed processing** with **intelligent multi-agent systems** to manage large-scale smart home sensor data efficiently. RDD transformations such as `map()`, `filter()`, and `reduceByKey()` enable fast parallel processing of temperature, motion, and energy data. Intelligent agents coordinate to optimize **comfort, energy efficiency, and security**, while continuously learning occupant behavior patterns. This architecture provides a scalable, adaptive, and efficient solution for managing thousands of connected smart homes.

RUBRICS FOR CALCULATION

Evaluation Criteria	Problem Understanding & Requirement Analysis	System Architecture Design	Use of PySpark & RDD Operations	Intelligent Agent Integration	Scalability & Distributed Computing Design	Data Flow & Processing Pipeline	Innovation & Creativity	Clarity, Presentation & Documentation
Weightage	10%	20%	15%	15%	10%	10%	10%	10%

Criteria	Excellent (5)	Good (4)	Average (3)	Poor (2)
Problem Understanding & Requirement Analysis	Clearly defines problem, objectives, and constraints	Mostly clear with minor gaps	Basic understanding	Unclear or incorrect
System Architecture Design	Complete, scalable, well-structured architecture with diagrams	Good design with minor issues	Basic design, lacks detail	Poor/incomplete

Use of PySpark & RDD Operations	Efficient and correct use of RDD transformations /actions	Mostly correct usage	Limited or partial use	Incorrect/missing
Intelligent Agent Integration	Strong integration with clear agent roles and logic	Good integration	Basic integration	Poor/no integration
Scalability & Distributed Computing Design	Clearly addresses scalability, fault tolerance, parallelism	Covers most aspects	Limited consideration	Not addressed
Data Flow & Processing Pipeline	Clear, logical, and well-defined data flow	Mostly clear	Some gaps	Poorly defined
Innovation & Creativity	Highly innovative, realistic, and impactful solution	Some innovation	Basic idea	No originality
Clarity, Presentation & Documentation	Very clear, neat diagrams, well-organized report	Mostly clear	Somewhat organized	Poor presentation